STAT 641 HW4 Hao Wang 28b878611

2.1 Ω_1, Ω_2 two nonempty sets. $T: \Omega_1 \to \Omega_2$

For any collection $\{A_\alpha : \alpha \in I\}$ of subsets of Ω_2.

① $T^{-1}(\bigcup_{\alpha\in I} A_\alpha) = \bigcup_{\alpha\in I} T^{-1}(A_\alpha)$

For any α, $A_\alpha \subset \bigcup_{\alpha\in I} A_\alpha \Rightarrow T^{-1}(A_\alpha) \subset T^{-1}(\bigcup_{\alpha\in I} A_\alpha) \Rightarrow \bigcup_{\alpha\in I} T^{-1}(A_\alpha) \subset T^{-1}(\bigcup_{\alpha\in I} A_\alpha)$

And,

For any element x in $T^{-1}(\bigcup_{\alpha\in I} A_\alpha)$, $x \in T^{-1}(\bigcup_{\alpha\in I} A_\alpha) \Rightarrow T(x) \in \bigcup_{\alpha\in I} A_\alpha$

$\Rightarrow$ there is a least one $\alpha_0 \in I$ s.t. $T(x) \in A_{\alpha_0} \Rightarrow x \in T^{-1}(A_{\alpha_0}) \subset \bigcup_{\alpha\in I} T^{-1}(A_\alpha)$

$\Rightarrow T^{-1}(\bigcup_{\alpha\in I} A_\alpha) \subset \bigcup_{\alpha\in I} T^{-1}(A_\alpha)$

$\Rightarrow T^{-1}(\bigcup_{\alpha\in I} A_\alpha) = \bigcup_{\alpha\in I} T^{-1}(A_\alpha)$

② $T^{-1}(\bigcap_{\alpha\in I} A_\alpha) = \bigcap_{\alpha\in I} T^{-1}(A_\alpha)$

$\forall x \in T^{-1}(\bigcap_{\alpha\in I} A_\alpha) \Rightarrow T(x) \in \bigcap_{\alpha\in I} A_\alpha \Rightarrow T(x) \in A_\alpha$ for any $\alpha \in I$.

$\Rightarrow x \in T^{-1}(A_\alpha)\ \forall \alpha \in I \Rightarrow x \in \bigcap_{\alpha\in I} T^{-1}(A_\alpha) \Rightarrow T^{-1}(\bigcap A_\alpha) \subset \bigcap T^{-1}(A_\alpha)$

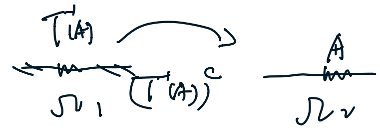
And,

$\forall x \in \bigcap_{\alpha\in I} T^{-1}(A_\alpha) \Rightarrow x \in T^{-1}(A_\alpha)$ for any $\alpha \in I \Rightarrow T(x) \in A_\alpha\ \forall \alpha \in I$

$\Rightarrow T(x) \in \bigcap_{\alpha\in I} A_\alpha \Rightarrow x \in T^{-1}(\bigcap_{\alpha\in I} A_\alpha) \Rightarrow \bigcap_{\alpha\in I} T^{-1}(A_\alpha) \subset T^{-1}(\bigcap_{\alpha\in I} A_\alpha)$

$\Rightarrow T^{-1}(\bigcap_{\alpha\in I} A_\alpha) = \bigcap_{\alpha\in I} T^{-1}(A_\alpha)$

③ $(T^{-1}(A))^c = T^{-1}(A^c)$



$\forall x \in (T^{-1}(A))^c \Leftrightarrow T(x) \notin A \Leftrightarrow T(x) \in A^c \Leftrightarrow x \in T^{-1}(A^c)$

2.3. $f, g: \Omega \mapsto \mathbb{R}$ be $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}) \rangle$-measurable functions. Set

$$h(\omega) = \frac{f(\omega)}{g(\omega)} I(g(\omega) \neq 0) \qquad \omega \in \Omega$$

$f^{-1}((-\infty, a]) \equiv \{\omega : f(\omega) \le a\} \in \mathcal{F}$; $g^{-1}((-\infty, a]) \equiv \{\omega : g(\omega) \le a\} \in \mathcal{F}$.

Let $h = h_2 \circ h_1 : \Omega \mapsto \mathbb{R} \Rightarrow h(\omega) = h_2(h_1(\omega))$

where $h_1(\omega) = (f(\omega), g(\omega))$ $h_2(f(\omega), g(\omega)) = \frac{f(\omega)}{g(\omega)} I(g(\omega) \neq 0)$

function h_2 is $h_2(x, y) = \frac{x}{y} I(y \neq 0)$

$h_2^{-1}((-\infty, a]) = \{(x,y) : \frac{x}{y} \le a, y \ne 0\} \cup \{(x,y)\ 0 \le a, y = 0\}$

$\Rightarrow \forall a \quad h_2^{-1}((-\infty, a]) \in \mathcal{B}(\mathbb{R}^2) \Rightarrow h_2 : \mathbb{R}^2 \to \mathbb{R}$ is $\langle \mathcal{B}(\mathbb{R}^2), \mathcal{B}(\mathbb{R}) \rangle$-measurable.

Also. by the proposition 2.1.3. $h_1(w) = (f(w), g(w))$ is $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}^2) \rangle$-measurable

$\Rightarrow h_2(h_1(w)) = h(w)$ is $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}) \rangle$ - measurable.

2.6. X_i, $i = 1,2,3$. r.v. on probability space $(\Omega, \mathcal{F}, P)$

$$\forall t \in \mathbb{R} \quad X_1(w) t^2 + X_2(w) t + X_3(w) = 0$$

(a) Show that $A \equiv \{w \in \Omega : \text{Equation has two roots}\} \in \mathcal{F}$.

$A = \{w : X_2^2(w) - 4X_1(w) X_3(w) > 0 \text{ and } X_1(w) \ne 0\}$

$h(x, y, z) = x^2 - yz$ is continuous.

$\Rightarrow \langle \mathcal{B}(\mathbb{R}^3), \mathcal{B}(\mathbb{R}) \rangle$ - measurable.

(b) $T_1(w)$ and $T_2(w)$ are two roots of the equation. on A.

$$f_i(w) = \begin{cases} T_i(w) & \text{on } A \\ 0 & \text{on } A^c \end{cases} \quad i = 1, 2.$$

Show that (f_1, f_2) is $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}^2) \rangle$ - measurable.

$$T_1(w) = \frac{-X_2 + \sqrt{X_2^2 - 4X_1X_3}}{2X_1}, \quad T_2(w) = \frac{-X_2 - \sqrt{X_2^2 - 4X_1X_3}}{2X_1}$$

$$\Rightarrow f_1(w) = T_1(w) \cdot I(w : w \in A)$$
$$= \frac{(-X_2 + \sqrt{X_2^2 - 4X_1X_3}\ \ I(w : X_2^2 - 4X_1X_3 > 0)}{2X_1} \ I(w : X_1 \ne 0)$$

$\Rightarrow h(x, y) = (x + \sqrt{y}) \cdot I(y > 0) = x \cdot I(y > 0) + \sqrt{y}\, I(y > 0)$ is $\langle \mathcal{B}(\mathbb{R}^2), \mathcal{B}(\mathbb{R}) \rangle$ - measurable.

$\Rightarrow$ By collary 2.1.4. $g_1(w) \equiv (-X_2 + \sqrt{X_2^2 - 4X_1X_3}) \cdot I(w : X_2^2 - 4X_1X_3 > 0)$ is $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}) \rangle$ - measurable.

And $g_2(w) = 2X_1(w) \Rightarrow f_1(w) = \frac{g_1(w)}{g_2(w)} I(w : g_2(w) \ne 0)$ is $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}) \rangle$ - measurable.

Similarly $f_2(w)$ is also $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}) \rangle$ - measurable.

Extra Problem.

A. $X: \Omega_1 \mapsto \Omega_2$ be a $\langle \mathcal{F}_1, \mathcal{F}_2 \rangle$ be a measurable transformation.

$$\sigma\langle X \rangle = \{ X^{-1}(B), B \in \mathcal{F}_2 \}.$$

Show (a) $\sigma\langle X \rangle$ is a σ-algebra.

X is $\langle \mathcal{F}_1, \mathcal{F}_2 \rangle$-measurable $\Rightarrow$ $\forall B \in \mathcal{F}$, $X^{-1}(B) \in \mathcal{F}_1$

$\Rightarrow$ $\sigma\langle X \rangle$ is a σ-algebra on Ω_1

(b) $\sigma\langle X \rangle$ is the smallest σ-algebra that makes X measurable.
(i.e. if X is $\langle \mathcal{G}, \mathcal{F}_2 \rangle$-measurable, then $\sigma\langle X \rangle \subset \mathcal{G}$).

For any σ-algebra $\mathcal{G}$ on Ω_1 s.t. X is measurable $\langle \mathcal{G}, \mathcal{F}_2 \rangle$-measurable.

$\forall B \in \mathcal{F}_2$, $X^{-1}(B) \in \mathcal{G}$ $\Rightarrow$ $\{ X^{-1}(B) = B \in \mathcal{F}_2 \} \subset \mathcal{G}$.

$\Rightarrow$ $\sigma\langle X \rangle \subset \mathcal{G}$.

B. For any set $A \notin \mathcal{B}(\mathbb{R})$, $f(x) = \begin{cases} 1 & x \in A \\ 0 & x \in A^c \end{cases}$ is not $\langle \mathcal{B}(\mathbb{R}), \mathcal{B}(\mathbb{R}) \rangle$-measurable.